

Katie Fitzsimons

The Pennsylvania State University
Mechanical Engineering
Human-Centered Robotics Lab

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EDUCATION

- Ph.D.*, Mechanical Engineering
Northwestern University, Evanston, IL 2020
Thesis: Motion as an Information Signal in Physical Human-Robot Interaction
Advisor: Todd D. Murphey, Ph.D.
- M.S.*, Mechanical Engineering
Northwestern University, Evanston, IL 2017
Thesis: Model-based Assistance for Human-in-the-loop Control
- B.S.*, Mechanical Engineering
Michigan State University, East Lansing, MI 2013

PUBLICATIONS

- [19] Evan Watts, Jonathon Slightam, Hershel Pangborn, **K. Fitzsimons**, and Ilya Kovalenko. Exploration and Navigation in a Partially Observable Cluttered Environment using Model Predictive Control and Heat Equation Driven Area Coverage. In IEEE Conference on Control Technology and Applications, Accepted.
- [18] Zachary Jester, Hongliang Li, **K. Fitzsimons**, Brian Zajac, and Ilya Kovalenko. Leveraging Digital Twins for Industrial Automation and Robotics Education. In ASEE Annual Conference and Exposition, Accepted.
- [17] Zachary Logan, Sophie Walsh, Quentin Anderson-Watson, and **K. Fitzsimons**. A proof-of-concept framework for adaptive haptic feedback via reinforcement learning. In IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics, 2026.
- [16] Junru Pang, Quentin Anderson-Watson, and **K. Fitzsimons**. Ergodic Imitation with Corrections: Learning from Implicit Information in Human Feedback. In IEEE Transactions on Human-Machine Systems, 2025.
- [15] Zachary Logan, Quinn Deitrick, and **K. Fitzsimons**. Evaluating factors affecting the perception of multi-sensory vibration and skin-squeeze cues during voluntary movement. In IEEE Transactions on Haptics, 2025.
- [14] Zhiqing Lu, Hershel Pangborn, and **K. Fitzsimons**. Developing an Affordable Multi-Course Electronics Kit for Increased Software and Mechatronic Literacy in Mechanical Engineering. In 2023 ASEE Zone 1 Conference, 2023.
- [13] **K. Fitzsimons** and T. D. Murphey. Ergodic shared control: Closing the loop on pHRI based on information encoded in motion. In ACM Tran. Hum.-Robot Interaction, 2022.
- [12] A. Kalinowska, A. Prabhakar, **K. Fitzsimons**, and T. D. Murphey. Ergodic imitation: Learning from what to do and what *not* to do. In *Int. Conf. on Robotics and Automation (ICRA)*, 2021.

- [11] A. Kalinowska, K. Rudy, M. Schlafly, **K. Fitzsimons**, J. Dewald and T. D. Murphey. Shoulder abduction loading affects motor coordination in individuals with chronic stroke, informing targeted rehabilitation. In *IEEE RAS/EMBS Int. Conf. on Biomedical Robotics and Biomechatronics (BioRob)*, 2020.
- [10] **K. Fitzsimons**, A. Kalinowska, J. P. Dewald, and T. D. Murphey. Task-based hybrid shared control for training through forceful interaction. *International Journal of Robotics Research* 39(9):1139–1154, 2020.
- [9] **K. Fitzsimons**, A. M. Acosta, J. P. Dewald, and T. D. Murphey. Ergodicity reveals assistance and learning from physical human-robot interaction. *Science Robotics*. 4(29), 2019.
- [8] T. Berrueta, A. Pervan, **K. Fitzsimons**, and T. D. Murphey. Dynamical system segmentation for information measures in motion. *Robotics and Automation Letters*, 4(1):169–176, 2018.
- [7] A. Kalinowska, **K. Fitzsimons**¹, J. P. Dewald, and T. D. Murphey. Online user assessment for minimal intervention during task-based robotic assistance. In *Robotics: Science and Systems*, 2018.
- [6] **K. Fitzsimons**, E. Tzorakoleftherakis, and T. D. Murphey. Optimal human-in-the-loop interfaces based on Maxwell’s Demon. In *American Control Conference (ACC)*, pages 4397–4402, July 2016.
- [5] B. T. Weaver, **K. Fitzsimons**, J. Braman, and R. Haut. The role of shoe design on the prediction of free torque at the shoe–surface interface using pressure insole technology. *Sports biomechanics*, 15(3):370–384, 2016.
- [4] K. Luchini-Collbry, T. Ortega-Knight, C. H. Chen, D. Lynch, **K. Fitzsimons**², C. D. Alton, J. L. M. Lapaix, J. Drost and G. Kohler. Exploring the experience of undergraduate research: a case study using Facebook. In *ASME Annual Conference & Exposition* (pp. 23-580), June 2013.
- [3] B. T. Weaver, **K. Fitzsimons**, J. E. Braman, and R. C. Haut. Torque prediction at the shoe–surface interface using insole pressure technology. *Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology*, 227(4):219–225, 2013.
- [2] B. T. Weaver, **K. Fitzsimons**, J. E. Braman, and R. C. Haut. The use of plantar insole pressure sensors to predict the free torque produced at the shoe-surface interface during internal rotation of the body relative to a planted foot. In *ASME Bioengineering Conference*, pages V01BT38A003–V01BT38A003. American Society of Mechanical Engineers, 2013.
- [1] K. D. Button, F. Wei, E. G. Meyer, **K. Fitzsimons**, and R. C. Haut. Determination of in situ ankle ligament strains in cases of high and medial ankle sprains. In *ASME Bioengineering Conference*, pages 275–276. American Society of Mechanical Engineers, 2012.

FUNDING

- Guha Maogharan (PI), Robert Voigt (co-PI), Vittaldas Prabhu (co-PI), Katie Fitzsimons(co-PI), and Craig Dubler (co-I) America Makes Impact 3.0: *Digi-FOCUS: Digital - Foundry Optimized for Cyber-physical and Agile Systems*, \$1,500,000, 10/2025-03/2027.

¹A. Kalinowska and K.Fitzsimons contributed equally to this work.

²K. Fitzsimons gave the oral presentation of this work.

- Rahman Azari(PI), Katie Fitzsimons (co-PI), Tamy Guimaraes (co-PI), Julian Wang (co-PI) and Lisa Iulo (co-PI) Coccoziello Institute for Real Estate Innovation:*DECODE: Digital twin Environments for COmprehensive Decarbonization Evaluation of the urban and rural real estate*, \$50,036, 07/2025-06/2026.
- Xiaogang Hu (PI), Huanyu Cheng (co-PI), Nanyin Zhang (co-PI), Katie Fitzsimons(co-PI), and Robert Sainburg (co-PI) National Science Foundation—Integrative Strategies for Understanding Neural and Cognitive Systems: *NSF-FR: Bidirectional Neural-Machine Interface for Closed-Loop Control of Prostheses*, \$3,999,570, 09/2023-08/2028.
- Katie Fitzsimons (co-PI) and Hershel Pangborn (co-PI) Leonhard Center for Enhancement of Engineering Education:*Entrance to Major Hardware Kit for Mechanical Engineering Undergraduates*, \$9,937, 05/2022-08/2023.

PROFESSIONAL SERVICE ACTIVITIES

Organizer

- National Workshop on Human-Like Robotics 2022
- Tutorial on Ergodic Control @ ICRA 2023-2024
- Workshop on Enabling Personalization in Assistive and Rehabilitation Robotics @ ICRA 2025-2026

Associate Editor

- IEEE Transaction on Human-Robot Interaction 2026-Present
- IEEE Transactions on Medical Robotics and Bionics 2023-Present
- International Conference on Robotics and Automation 2022, 2023, 2025

Reviewer

- IEEE Transactions on Automation Science and Engineering 2025,2026
- IEEE Transaction on Medical Robotics and Bionics 2023, 2025
- International Journal of Human-Computer Studies 2025
- IEEE Transaction on Human-Machine Systems 2021, 2023, 2024, 2025,2026
- International Conference on Robotics and Automation 2022, 2023, 2024, 2025
- IEEE Robotics and Automation Letters 2021, 2024, 2025
- Conference on Robot Learning 2021, 2022,2023, 2024
- IEEE Haptics Symposium 2024
- International Conference on Intelligent Robots and Systems 2021, 2023
- ACM Transactions on Human-Robot Interaction 2021, 2022
- IEEE Conference on Decision and Control 2022
- IEEE-RAS International Conference on Humanoid Robots 2022
- IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics 2020
- Robotics: Science and Systems 2019,2026

Penn State

- Schreyer Honors Advisor 2025-Present
- Judge for Engineering Summer Bridge Program Robotics Competition 2025
- Faculty Selection Committee for the Millenium Scholars Program 2024, 2025
- Faculty Selection Committee for the Schreyer Honors College 2022, 2023, 2025
- Human Powered Vehicle Team Faculty Sponsor 2021-Present
- Robot in 3 Days (Ri3D) Faculty Sponsor 2023-Present
- ME Faculty Search Committee 2021-2022

Panelist

- Subject Matter Expert for National Defense Science and Engineering Graduate Fellowship 2021,2022,2026
- National Science Foundation 2021, 2022, 2023, 2024, 2025, 2026

Reviewer for Swiss National Science Foundation 2026
Peer Review for Humboldt Research Fellowship 2025
Meta-Reviewer for RSS Pioneers 2021
IEEE, Member 2015-Present

K-12 Outreach

Centre County 4H Robotics Team 2023-Present

- Developed and delivered a workshop on design thinking to FRC (high school) robotics team;
- Mentored First Lego League team each week from August to February competition.

Pittsburgh Robotics Discovery Day 2023; 2024; 2025

- Coordinated Human-Centered Robotics lab demonstrations;
- Presented work on haptics and robotics to hundreds of middle and high school students at the event.

Pennsylvania FIRST Technical Challenge 2025

- Volunteered as judge for the state championship competition

Museum of Science and Industry Robot Block Party Spring 2013; 2016; 2017; 2018

- Organized the transport and maintenance of equipment for demonstrations during National Robotics Week;
- Presented work on haptics and robotis two thousand of visitors at Chicago's MSI;
- Discussed past and current research project with children and adults.

Northwestern University STEM & Sports Event Fall 2016; 2017

- Coordinated with 3 other research groups and Northwestern athletics department to arrange tours and demonstrations;
- Let two tour groups of 50+ elementary school children and chaperones through the Neuroscience and Robotics Lab.

GradSWE Outreach Event for High school students

Fall 2018

- Hosted a group of 16-18 year old girls from Chicago public schools at Northwestern University
- Prepared an oral presentation about my work and my past experiences;
- Discussed college and STEM careers with small groups of 4-5 students.

INVITED TALKS

An Information-Theoretic Approach to Evaluation and Control of Human-Robot Motion.

University of Minnesota, MN Feb. 2019

Michigan State University, MI April 2019

Rethinking Motion Measures for Physical Human-Robot Interaction

Villanova University, PA Jan. 2020

Pennsylvania State University, PA Feb. 2020

University of Texas, TX Feb. 2020

University of Washington, WA March 2020

Harvard University, Virtual March 2020

University of Louisville, KY March 2020

University of Massachusetts, Virtual March 2020

Using Information Encoded in Motion to Close the Loop on pHRI.

ASME International Design Engineering Technical Conference: SEC-session, August 2021

Using Information Encoded in Motion to close the loop in pHRI

Penn State Action Club, February 2022

Technical Introduction to Ergodic Control

Tutorial on Ergodic Control @ ICRA, May 2024

Learning through Physical Human-Robot Interactions.

Penn State BioRobotics Symposium, February 2025

Learning through Physical Human-Autonomy Interactions.

Penn State College of Medicine Physical Medicine & Rehabiliations Scientific Day, April 2025

Neuromotor Learning Supported by ‘Random’ Robot Actions.

Penn State Center for Neural Engineering Seminar Series, December 2025

Conference Presentations

- [1] N. M. Kitchen, Z. Logan, **K. Fitzsimons**, X. Hu and R. L. Sainburg. Sensorimotor adaptation as a means of testing robustness of human-robot interfaces in neuroprosthetics. In *RehabWeek*, Society of Neuro Mechanics in Rehab, 2025.

PROFESSIONAL DEVELOPMENT

- Practical AI for Engineering Courses: Creating Homework and Evaluating Student Learning April 2026
- COE Undergraduate Advising Workshop Part 2 February 2026
- Engaging Classroom Examples! Using demonstrations and real-life examples to encourage fun and engagement with your students October 2025
- Integrating AI into the Undergraduate Mechanical Engineering Curriculum: Pedagogy and Assessment October 2025
- COE Undergraduate Advising Workshop September 2025
- Penn State TopHat Days September 2024
- Leonhard Center Workshop: Developing an Inclusive Syllabus December 2023
- Leonhard Center Workshop: Remembering Name and Faces August 2023
- Leonhard Center Workshop: Using Kahoot to Teach Engineering Students November 2021

STUDENTS ADVISED

PhD Students

- Zachary Logan, 2026 Ph.D. thesis: Task-Based Haptic Cueing for Assistance and Training
- Junru Pang, Expected 2026 Ph.D. Thesis: Online Adaptation for Human-Guided Robot Learning Framework: from Error Responding to User Style Generalization
- Quentin Anderson-Watson, Expected 2029
- Arnab Gupta, Expected 2029
- Oluwatosin Ajibola, Expected 2030

Master's Students

- Zhiqing Lu, 2023.
- Ryan Tracy, 2025. (Penn State World Campus)
- Savanna Spazak, Expected 2026
- Quinn Deitrick, 2025. National Science Foundation Graduate Research Fellow